MGCR-Net: Multimodal Graph-Conditioned Vision-Language Reconstruction Network for Remote Sensing Change Detection

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***Abstract*—With the advancement of remote sensing satellite technology and the rapid progress of deep learning, remote sensing change detection (RSCD) has become a key technique for regional monitoring. Traditional change detection (CD) methods and deep learning-based approaches have made significant contributions to change analysis and detection, however, many outstanding methods still face limitations in the exploration and application of multimodal data. To address this, we propose the multimodal graph-conditioned vision-language reconstruction network (MGCR-Net) to further explore the semantic interaction capabilities of multimodal data. Multimodal large language models (MLLMs) have attracted widespread attention for their outstanding performance in computer vision, particularly due to their powerful visual-language understanding and dialogic interaction capabilities. Specifically, we design a MLLM-based optimization strategy to generate multimodal textual data from the original CD images, which serve as textual input to MGCR. Visual and textual features are extracted through a dual encoder framework. For the first time in the RSCD task, we introduce a multimodal graph-conditioned vision-language (VL) reconstruction mechanism, which is integrated with graph attention to construct a semantic graph-conditioned reconstruction module (SGCM), and this module generates VL tokens through graph-based conditions and enables cross-dimensional interaction between visual and textual features via multihead attention. The reconstructed VL features are then deeply fused using the language vision transformer (LViT), achieving fine-grained feature alignment and high-level semantic interaction. Experimental results on four public datasets demonstrate that MGCR achieves superior performance compared to mainstream CD methods. Our code is available on https://github.com/cn-xvkong/MGCR**

Received 15 June 2025; revised 9 October 2025 and 4 November 2025; accepted 12 January 2026. Date of publication 15 January 2026; date of current version 29 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62301105, Grant U24A20219, and Grant 62272281; in part by Taishan Scholars Project of Shandong Province under Grant tsqn202306274 and Grant tsqn202507240; in part by the Natural Science Foundation of Shandong Province under Grant ZR2025QC712, Grant ZR2025MS985, and Grant ZR2025QC695; and in part by Yantai Science and Technology Innovation Development Plan under Grant 2023JCYJ040 and Grant 2022JCYJ031. *(Corresponding author: Guodong Fan.)*

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Digital Object Identifier 10.1109/TGRS.2026.3654629

***Index Terms*—Graph-conditioned vision-language (VL) reconstruction, language vision transformer (LViT), multimodal artificial intelligence, remote sensing change detection (RSCD).**

I. Introduction

REMOTE sensing change detection (RSCD) focuses on identifying changes in ground objects from bitemporal images and has become an inevitable outcome of remote sensing advancements aimed at achieving regional monitoring. Specifically, this technology can be applied to land use statistics, disaster impact assessment, and forest cover mapping, among other tasks [1], [2], [3]. In this work, we focus on binary change detection (CD) to construct MGCR, where the prediction results are presented as binary classification maps. Traditional CD methods include pixel-level, object-level, and scene-level CD [4], [5], [6], which have demonstrated strong performance in manually labeled change analysis but require substantial resources and suffer from limited adaptability. Machine-learning approaches learn from raw data and extract features to adaptively optimize model weights. Algorithms such as support vector machines, random forests, and k-means clustering [7], [8], [9], [10] have been employed to perform binary classification and detect changes between bitemporal images from different perspectives. However, they still exhibit limitations in capturing deep semantic information.

The development of deep learning methods has effectively addressed the limitations of traditional machine learning in CD tasks by significantly simplifying the complex feature engineering processes involved in conventional CD pipelines, while also reducing human intervention and computational costs. Convolutional neural networks (CNNs), with their strong local perception capabilities and parameter-sharing mechanisms, can efficiently extract local features from images, enabling accurate pixel-level classification and precise localization of change regions. However, CNNs are inherently limited in modeling long-range dependencies and struggle to capture global correlations between nonlocal features in images. The transformer architecture [11] effectively addresses this issue by using self-attention to capture long-range dependencies and extract global contextual information from the entire image. The vision transformer (ViT) [12] introduces the transformer architecture into computer vision, enabling global image modeling through patch embedding. Building on this,

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the pyramid vision transformer (PVT) [13] enhance multiscale representation and computational efficiency. Extending beyond visual tasks, language vision transformer (LViT) [14] integrates image and text modalities within a unified framework, leveraging semantic information to guide visual understanding and mitigate limitations in image quality. Recently, numerous innovative CD methods based on CNNs and transformers have emerged [15], [16], [17], [18], [19], achieving impressive results in addressing various challenges. Many methods and architectures still provoke our reflection [20], [21], [22], [23], [24], [25], [26], [27], [28], [29]. However, most existing methods emphasize structural or pixel-level differences while neglecting the complex nonlinear and semantic relationships in multimodal remote sensing data. To address this, we propose a text-guided multimodal model that enhances visual-linguistic understanding and interaction.

In recent years, visual-language representation learning has achieved remarkable success, driving rapid advancements in image-text understanding within the field of computer vision. With the evolution of model architectures and pretraining strategies, tasks such as image captioning [30], text-to-image generation [31], visual question answering [32], and visual reasoning [33] have flourished, becoming central topics in multimodal research. Among them, contrastive learning has been widely adopted for vision-language (VL) modeling. For instance, CLIP [34] learns to embed images and texts into a shared semantic space through large-scale image–text pair training, enabling powerful zero-shot recognition capabilities. ALIGN [35] builds upon this by leveraging massive web-crawled image–text data to reduce reliance on manual annotations. LLaVA [36] integrates CLIP with Vicuna to construct a multimodal system capable of engaging in visual question answering. Despite their impressive performance in handling nonlinear and complex image–text data, these models still face limitations in scenarios like RSCD, which demand deeper semantic understanding and cross-temporal modeling, highlighting gaps in their generalizability and expressive power.

Thus, we propose MGCR, designed for semantic-level change perception in bitemporal remote sensing images. Specifically, we utilize the number and spatial distribution of buildings as prior cues to guide a multimodal captioning strategy based on the LLaVA model, generating structured textual descriptions from bitemporal imagery, as illustrated in Fig. 1. To avoid overly verbose and complex outputs, we incorporate a regular expression-based semantic pruning strategy to extract concise, change-relevant textual content. MGCR adopts a weight-sharing Siamese architecture and leverages PVT to extract visual features from the bitemporal image pair, while semantic features are derived from the corresponding texts using CLIP's text encoder. In the VL alignment stage, we introduce a graph-based conditional reconstruction module to deeply model the cross-modal semantic dependencies between image and text. This module maps multimodal features into a heterogeneous graph structure, where contextual feature interaction is achieved by constructing dependency edges between nodes. In this graph, nodes represent semantic units from different modalities, and edge weights dynamically capture

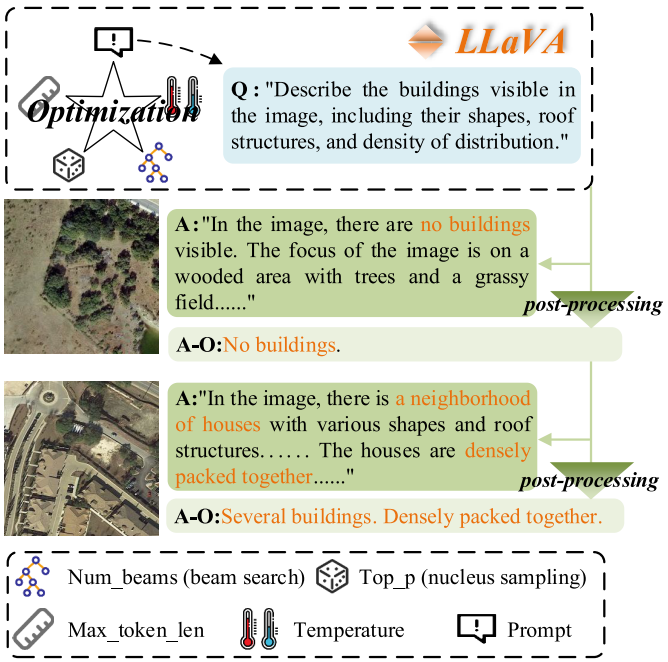


Fig. 1. Illustrates our optimization strategy for LLaVA. By designing tailored prompts and tuning multiple parameters, we guide the model to generate text descriptions relevant to the input images.

cross-modal relevance, enabling both explicit alignment and implicit fusion. Finally, we integrate LViT to deeply fuse the reconstructed multimodal representations, using its hierarchical interaction mechanism to perform layered modeling and semantic refinement, thereby enhancing the model's ability to perceive semantic changes. The main contributions of this article are as follows.

1) We optimize the text generation strategy of LLaVA to produce concise and image-relevant descriptions for CD datasets, and we construct a unified multimodal CD framework MGCR for deep semantic fusion between images and text.
2) We propose a graph-based conditional reconstruction module to model semantic dependencies between visual and textual modalities, reconstructing fused features through graph attention, and further employ LViT for deep multimodal fusion to enhance semantic understanding.
3) We conduct comprehensive evaluations of MGCR on four benchmark RSCD datasets and compare it against a variety of representative methods, achieving superior performance across multiple evaluation metrics.

II. Related Work

A. Remote Sensing CD Methods

With advances in remote sensing, high-resolution imagery has accelerated the development of CD methods. Traditional approaches, ranging from pixel-level differencing to feature-based extraction, explored various directions but were limited by high computational costs. Deep learning has mitigated these issues, and models such as FC-EF, FC-Siam-conc, and FC-Siam-diff [37], which are based on fully convolutional and

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Siamese architectures, have become fundamental frameworks for modern CD through different feature fusion strategies. IFNet [38] extracts deep features via a fully convolutional structure and integrates a differential discriminator with attention mechanisms for image reconstruction. SNUNet [39] introduces a densely connected Siamese network to alleviate deep feature loss via compact information transmission, and it incorporates channel attention modules for deep supervision. BIT [40] employs a transformer backbone to model spatiotemporal contextual information, feeding it back into the pixel space and refining original features through a decoder. DTCDSCN [41] supports both CD and semantic segmentation, incorporating dual attention modules to enhance feature representation. ChangeFormer [42] synergistically combines a transformer encoder with a multilayer perceptron decoder to effectively capture multiscale, long-range dependencies. ICIFNet [43] integrates CNNs and transformers to achieve intra-scale cross interaction and inter-scale fusion, tapping into the full potential of hybrid models. DMINet [44] unifies self-attention and cross-attention into joint attention mechanisms, enhancing global feature distribution, feature coupling, and difference detection. AERNet [45] constructs an attention-guided edge refinement network that strengthens feature learning via global aggregation, attention guidance, and an adaptive loss function. SEIFNet [46] introduces a spatiotemporal difference enhancement module to highlight change regions and uses an adaptive context fusion module to guide inter-layer semantic interactions. ChangeCLIP [47] reconstructs original CLIP-derived features, aligns visual and textual modalities, and builds a differential compensation module to capture fine-grained semantic changes.

B. Development and Applications of MLLMs

In recent years, LLM have expanded artificial intelligence from language processing to multimodal understanding and generation. Built on the transformer architecture, they learn semantic structures from large-scale corpora and, through instruction tuning or multitask learning, adapt to diverse tasks such as question answering, summarization, and dialog [48], [49], [50]. In multimodal VL tasks, CLIP overcomes the limitations of fixed-label image classification by using contrastive learning on large-scale image–text pairs. It maps images and texts into a shared semantic space through separate encoders, aligning semantically related pairs with high similarity. This cross-modal alignment enables open-vocabulary image retrieval and image–text matching, greatly improving model generality and scalability [51], [52], [53]. Building upon this, LLaVA integrates CLIP's visual perception capabilities with the language understanding and generation abilities of LLM like Vicuna, creating a powerful multimodal dialog system [54], [55]. LLaVA first extracts image features via CLIP, then transforms them into a language-understandable format using a visual projection module, allowing the LLM to generate natural language outputs based on complex visual-textual inputs. This architecture not only achieves effective alignment between image and text information but also provides good extensibility, supporting multimodal tasks such as multi-turn dialog and scene understanding. Furthermore, LLaVA's generative potential has been preliminarily validated, with researchers combining it with diffusion models for image–text generation tasks, expanding its application in the field of multimodal generation [56], [57]. Inspired by this, we incorporate the same principles and develop a LLaVA-based text generation optimization strategy to generate relevant textual descriptions from bitemporal remote sensing imagery.

C. Attention Mechanism

In CD tasks, attention mechanisms help models focus on differing regions in bitemporal remote sensing images, enhancing the extraction of critical information. Channel attention highlights key semantic channels, while spatial attention guides focus toward potential change areas and suppresses background noise. Their combination jointly optimizes content and positional features for more effective change representation [58], [59]. To improve alignment between bitemporal images, cross-attention has been introduced, using features from one image as queries and from the other as keys and values to establish precise semantic correspondences [60], [61]. Graph attention mechanisms [62], [63] model pixels as nodes in a graph structure, where adjacency matrices are computed to reinforce adaptive interactions between nodes, with edges representing relationships between regions. By dynamically allocating attention weights across nodes via the adjacency matrix, graph attention allows adaptive information exchange among features, thereby highlighting potential change regions and strengthening their contextual dependencies. Inspired by cross-attention, we construct the semantic graph-conditioned reconstruction module (SGCM) to reconstruct semantically aligned information from image–text pairs, generating VL Tokens as graph condition nodes. These tokens serve as keys and values, with image and text features as queries. A multihead attention mechanism is then employed to complete the multimodal interaction of image–text features.

III. Methodology

A. Overall Architecture

In this article, we propose an innovative method, MGCR, which leverages multimodal large language model (MLLM) to focus on the CD task. As shown in Fig. 2, our model is composed of multiple components: LLaVA for generating textual data, PVT and CLIP as image–text encoders, SGCM for conditional reconstruction and alignment between modalities, and a LViT-based structure for deep fusion of visual and textual features. First, we utilize LLaVA's image–text understanding and powerful user interaction capabilities to simulate a dialog process in which it responds to prompts about image content. From the extensive textual descriptions generated, we apply regular expression-based pruning to retain only text related to building count and spatial distribution, thereby constructing multimodal input data specifically tailored for building CD. Next, we employ PVT and CLIP models as encoders to extract features, using PVT's pyramid architecture to derive multiscale features from bitemporal images, while feeding the generated text into CLIP's text encoder to obtain textual representations. To effectively combine image and text features, we

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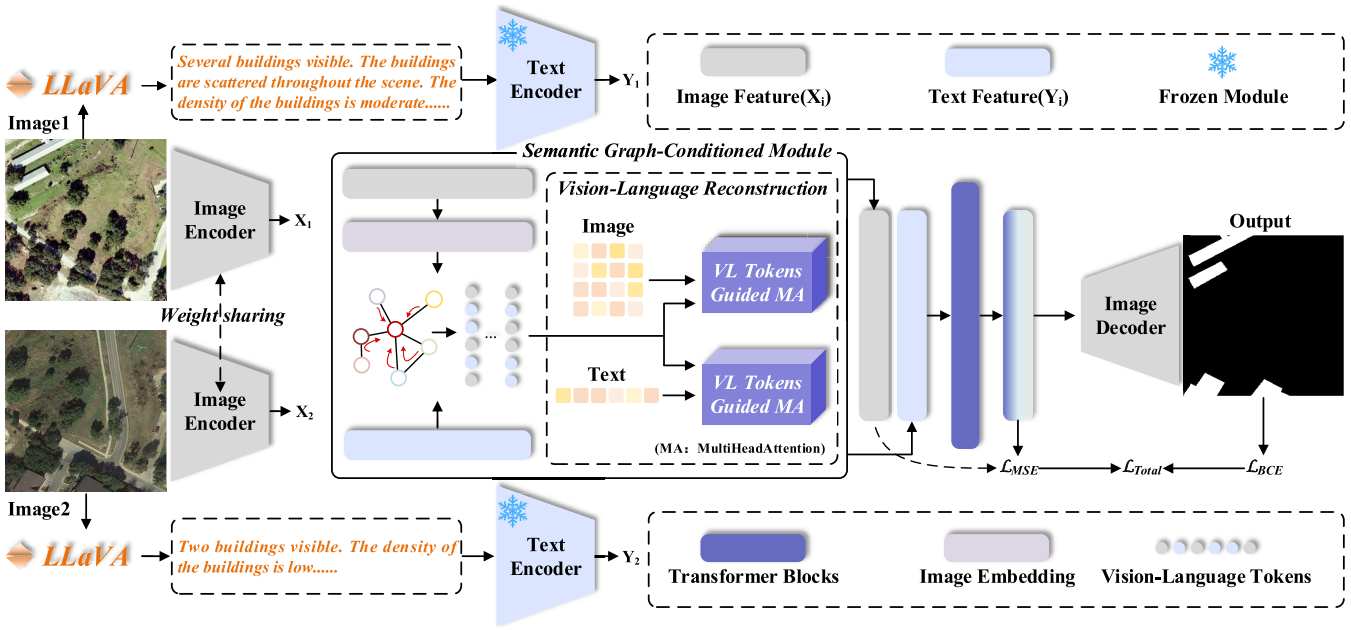


Fig. 2. Overall architecture of MGCR-Net is shown. The optimized textual data generated by LLaVA is encoded using the CLIP text encoder and fused with the visual features extracted by the PVT encoder. Subsequently, these multimodal features are reconstructed through the SGCM and LViT modules. Specifically, the SGCM employs a graph-conditioned attention mechanism to construct VL Tokens for image and text reconstruction, while the LViT module integrates the reconstructed visual and textual information, thereby achieving cross-dimensional semantic information interaction.

design SGCM based on graph attention, which models the correspondence between image regions and text tokens. This module treats salient image regions as graph nodes and uses textual guidance to construct adjacency relations, enabling structured interactions and feature reconstruction under semantic alignment, thus overcoming the limitations of unimodal CD by leveraging multidimensional information. Finally, in the cross-modal feature integration stage, we incorporate the LViT structure to perform deep fusion of image and text features. Through hierarchical self-attention, this structure dynamically learns semantic associations between modalities, enhancing the model's ability to discern building edges and morphological differences, and providing fine-grained semantic information.

B. Text Generation Based on Multimodal Large Models

In this article, we leverage the powerful image–text alignment and language modeling capabilities of LLaVA as a multimodal text generator. After task-specific optimization of LLaVA for building CD, we input the raw images, and LLaVA generates descriptive text containing structured semantic information, including the number of buildings, their spatial distribution, and density. Additionally, it provides judgments and descriptions of vehicles, roads, grass, trees, and other elements. The simplified mathematical formulation is as follows:

$$V = \mathcal{E}_v(X) \tag{1}$$

$$V' = W_{pV} + b_p \tag{2}$$

$$\hat{y}_t = \text{MLLM}\left(\text{Concat}\left(V', T_{\text{prompt}}\right)\right) \tag{3}$$

where X represents the raw image input, $\mathcal{E}_v$ denotes the visual encoder of CLIP, and extracts the visual features of the image to obtain V. W_p is a trainable linear projection matrix, and b_p is a trainable bias term. By linearly mapping the visual features and aligning them with the feature space of the language model, we obtain V', which is then integrated with T_{prompt} and passed through the MLLM to generate the predicted output $\hat{y}_t$.

C. Feature Encoder

PVT is a transformer architecture designed for computer vision tasks, with its core advantage lying in the introduction of a pyramid structure that endows the transformer with multiscale feature extraction capabilities similar to CNNs. PVT divides the input image into smaller patches and progressively reduces spatial resolution while enhancing semantic features layer by layer, achieving hierarchical feature representation. Each scale consists of multiple stages, where a spatial-reduction attention mechanism is employed to effectively reduce the computational cost of self-attention, enabling PVT to efficiently handle high-resolution images while preserving the long-range modeling capabilities of transformers. Ultimately, PVT outputs multiresolution pyramid features, offering rich multiscale semantic representations, and its mathematical formulation is as follows:

$$X_{ij} = E_i(X_i) \tag{4}$$

where E denotes the PVT encoder, i represents the Siamese network processing the bitemporal images, and j refers to the different scales obtained after feature extraction by the PVT encoder.

CLIP is a general VL pretraining architecture based on cross-modal contrastive learning. Its core idea is to encode images and natural language texts into a shared semantic embedding space by maximizing the similarity

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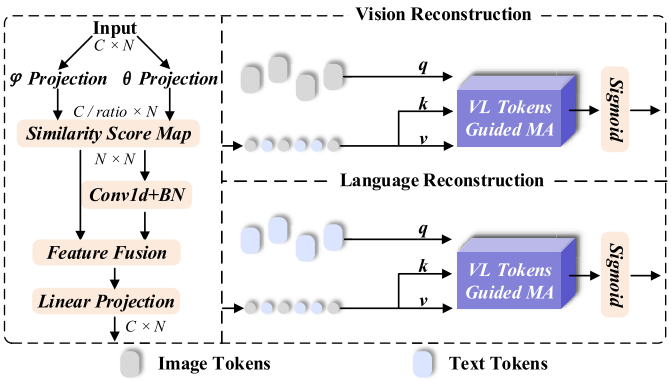


Fig. 3. Schematic illustrates the basic structure of SGCM, where a graph attention mechanism is introduced to facilitate interaction between image tokens and text tokens. Subsequently, VL Tokens are constructed and used separately for vision and language reconstruction. The reconstructed information is then deeply fused through the LViT module, enabling the model to effectively absorb multimodal information.

between matching image–text pairs and minimizing the similarity between mismatched pairs, thereby achieving semantic alignment across modalities. Unlike traditional classification models that rely on manually annotated labels, CLIP is trained on large-scale image–text pairs without predefined category labels, allowing it to automatically capture deep semantic relationships between visual and linguistic information and exhibit strong open-vocabulary understanding capabilities. In this article, we fully leverage CLIP's strengths in cross-modal modeling, specifically adopting its text encoder to transform natural language prompts into semantic vectors that can be aligned with visual features. This encoder not only comprehends word-level semantics but also captures sentence-level contextual structures, providing fine-grained modulation signals for image semantics. Compared to conventional visual encoders, CLIP's text embeddings carry rich linguistic priors learned during training, enabling the model to understand targets and relationships in images based on flexible natural language descriptions rather than being restricted to a fixed label set. This offers stronger semantic support for subsequent multimodal fusion and CD tasks. The corresponding mathematical formulation is as follows:

$$T_i = \mathcal{E}_{\text{ti}}(\hat{y}_{\text{ti}}) \tag{5}$$

where $\mathcal{E}_{\text{ti}}$ denotes the text encoder of CLIP, and T_i represents the text features obtained by encoding the textual descriptions of the bitemporal images using the text encoder.

D. Semantic Graph-Conditioned Module

In multimodal CD tasks, the effective integration of visual and textual features is crucial. Misalignment between visual and textual features can negatively impact the model's performance. Cross-attention-based fusion is a commonly used approach to align visual and textual features in multimodal tasks. The SGCM consists of three components, as shown in Fig. 3, first, graph attention is applied to achieve an initial fusion of visual and textual features, resulting in VL Tokens. Then, two graph-conditioned reconstruction modules are constructed—one for text-to-vision and the other for vision-to-text. These modules further enhance the semantic association between vision and language by automatically learning to reconstruct multimodal features during training

$$E_{X_{i3}} = \text{Embedding}(X_{i3}) + E_{\text{pos}} \tag{6}$$

$$\text{VL} = \text{Concat}\left(E_{X_{i3}}, T_i\right) \tag{7}$$

we adopt a single-layer interaction feature alignment strategy, where the fusion and alignment of visual and textual features are built upon the deep pyramid structure with global attribute features. Therefore, we select X_{i3} as the image input and apply an embedding operation along with the addition of positional encoding E_{pos} to achieve an initial structural alignment between the image and text. Subsequently, the aligned features are concatenated to obtain VL

$$\Phi = \text{LayerNorm}(\phi(\text{VL})) \tag{8}$$

$$\Theta = \text{LayerNorm}(\theta(\text{VL})) \tag{9}$$

$$z = \Phi \cdot \Theta^T \tag{10}$$

$$\mathcal{Z} = \text{BN}\left(\text{Conv1D}\left(z^T\right)\right)^T + z \tag{11}$$

and first, the fused representation VL undergoes two linear transformations to obtain Φ and Θ, which represent the relationships between node pairs. Then, an adjacency matrix z is constructed along the sequence dimension to model the similarity score map between visual and textual features. By integrating the graph concept with convolution operations, the overall graph of key VL nodes $\mathcal{Z}$ is constructed

$$F = \text{Concat}(z, \mathcal{Z}) \tag{12}$$

$$F' = \text{BN}\left(\text{Conv1D}\left(F^T\right)\right)^T \tag{13}$$

$$\text{VL Tokens} = \text{ReLU}\left(\text{VL} + \text{Proj}\left(F'\right)\right) \tag{14}$$

finally, the VL Tokens are projected into the feature space

$$K = V = \text{VL Tokens} \tag{15}$$

$$\text{VR} = \text{Softmax}\left(\frac{Q_{\text{img}}K^T}{\sqrt{d}}\right)V \tag{16}$$

$$\text{LR} = \text{Softmax}\left(\frac{Q_{\text{text}}K^T}{\sqrt{d}}\right)V \tag{17}$$

where VR denotes vision reconstruction and LR denotes language reconstruction. We employ a cross-attention mechanism to build semantic associations between the two modalities, using VL tokens as the keys and values, while using the image and text features, respectively, as the queries. This mechanism enables visual and textual features to extract contextual information from VL Tokens autonomously. By enhancing image representations via textual guidance and refining textual representations through visual cues, the model facilitates a profound collaborative semantic modeling of both vision and language modalities.

$$X'_{i3} = X_{i3} + \text{Sigmoid}(\text{VR}) \tag{18}$$

$$T'_i = T_i + \text{Sigmoid}(\text{LR}) \tag{19}$$

$$\text{VL} = \text{LViT}\left(X'_{i3}, T'_i\right) \tag{20}$$

Specifically, after the initial alignment and reconstruction of VL features, we feed them into a multilayer transformer

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encoder module to provide a richer and more semantically consistent representation for subsequent CD.

IV. Experimental Results and Analysis

Our optimized text generation data is specifically designed for building CD. To further verify the effectiveness and generalization ability of MGCR in broader multimodal CD tasks, we conduct experiments on publicly available datasets not limited to building CD, including LEVIR-CD [64], WHU-CD [65], GZ-CD [66], and additionally SYSU-CD [67]. We perform comparative experiments with several mainstream methods as well as ablation studies on MGCR to comprehensively evaluate the performance of the proposed model.

A. Datasets

1) LEVIR-CD Dataset: The LEVIR-CD dataset is specifically designed for building CD across diverse environments, including urban and suburban scenes. It consists of 637 image pairs, each with a resolution of 1024 $\times$ 1024 pixels and annotated at the pixel level. We randomly crop the original images into 256 $\times$ 256 patches and split them into training, validation, and test sets with a ratio of 7:2:1. This dataset features multiscale changes, diverse surface cover types, and complex backgrounds, making it suitable for evaluating the generalization and robustness of the model.

2) WHU-CD Dataset: The WHU-CD dataset focuses on building CD in the context of urban expansion. It includes various land cover types such as buildings, roads, and green areas, with fine-grained pixel-level annotations. The images are cropped into 256 $\times$ 256 patches, yielding a total of 7620 image pairs, which are split into training, validation, and test sets at a ratio of 8:1:1. This dataset is suitable for assessing the model's adaptability to different urban textures and surface change patterns.

3) GZ-CD Dataset: The GZ-CD dataset targets building CD in densely populated urban areas. It contains tasks involving both large-scale structures and small scattered buildings, while also presenting complex cases such as building shadows and overlapping regions. The images are cropped into 256 $\times$ 256 patches, with 2834 pairs for training, 400 for validation, and 325 for testing. This dataset is valuable for evaluating the model's ability to detect subtle changes in complex urban scenarios.

4) SYSU-CD Dataset: Released by Sun Yat-sen University, the SYSU-CD dataset is a high-resolution RSCD dataset featuring well-aligned image pairs and accurate labels. It includes typical urban building changes as well as other types of changes such as vegetation shifts and road expansion. Each image pair is sized 256 $\times$ 256, and the dataset contains a total of 20 000 pairs, split into training, validation, and test sets in an 8:1:1 ratio.

B. Implementation Environment

The experiments were conducted on an Ubuntu 18.04 system for code execution and debugging. Model development and training were performed using the PyTorch deep learning framework, accelerated by an NVIDIA TITAN RTX 24GB GPU. During the optimization of LLaVA's strategy, the following parameters were set, temperature = 0.1, top_p = 0.3, num_beams = 5, and max_new_tokens = 256. We provided prompts focused on building-related content to generate rich multimodal text data. For other types of CD tasks, such as SYSU-CD, we adopted a text data generation method similar to ChangeCLIP, using the probabilities of multiple background categories as text inputs. AdamW was selected as the optimizer with a learning rate of $5e^{-4}$ and a weight decay of 0.01. A cosine annealing schedule was applied for learning rate adjustment. The model training was limited to a maximum of 300 epochs. For guiding model optimization, we combined binary cross-entropy loss and mean-squared error loss. The complete loss structure can be expressed as

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{21}$$

$$L_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{22}$$

$$L_{\text{Total}} = \lambda_1 L_{\text{BCE}} + (\lambda_2 + \lambda_3) L_{\text{MSE}} \tag{23}$$

where λ_1, λ_2, and λ_3 are weighting hyperparameters. L_{BCE} supervises the MGCR-generated predicted maps against GT, while L_{MSE} measures the similarity between the features of the bitemporal images and the fused multimodal features, enhancing the consistency of the multimodal feature representations.

C. Evaluation Metrics

To evaluate the performance of our model on the CD task and compare it with other methods, we employ four commonly used metrics, $F1$, IoU, precision, and recall. The $F1$, being the harmonic mean of precision and recall, is especially suitable for imbalanced class scenarios and provides a more comprehensive performance assessment. IoU measures the spatial overlap between the predicted change regions and the ground truth change labels, reflecting the detection accuracy of the model. Precision indicates the proportion of correctly predicted positive samples among all predicted positives, while recall measures the model's ability to identify all true positive samples. The specific mathematical formulas for these metrics are provided below

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{24}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{25}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{26}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{27}$$

where TP represents true positives, FP represents false positives, FN represents false negatives, and TN represents true negatives.

D. Comparative Experiment

In this section, we evaluate the effectiveness of MGCR on four CD datasets: LEVIR-CD, WHU-CD, GZ-CD, and

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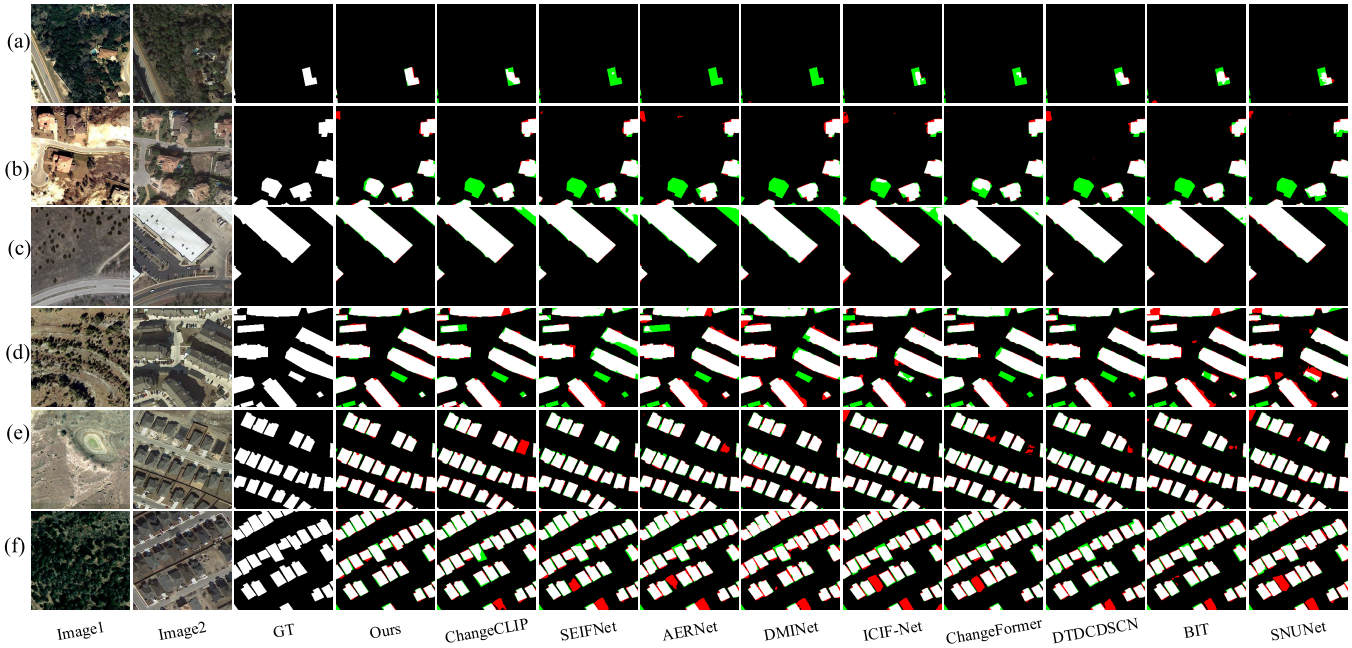


Fig. 4. Comparison of MGCR-Net with other methods, where (a)–(f) are representative examples from the LEVIR-CD dataset. In this figure, the green areas represent the predicted parts that are missing compared to the GT, while the red areas represent the redundant predicted parts compared to GT. The generation of specific textual information enables SGCM to capture buildings that other methods fail to predict through a vision-language reconstruction mechanism. The deep fusion of visual and textual information in LViT enhances the model's focus on change regions and buildings, effectively eliminating redundant predictions commonly observed in other models.

SYSU-CD. Throughout the comparison with other mainstream methods, MGCR consistently achieves the best performance both in terms of visualized prediction maps and commonly used quantitative metrics.

As shown in Fig. 4, we selected representative samples A–F from the LEVIR-CD dataset based on characteristics such as building quantity and density, choosing different perspectives for detailed analysis. Samples A and B have backgrounds dominated by roads and trees. In sample A, buildings surrounded by trees with colors similar to the foliage were difficult for many mainstream methods to distinguish as change targets. ChangeCLIP also showed imprecise detection of difference regions due to blurred boundaries between buildings and trees. In sample B, the building in the lower left corner blends into the surrounding background, causing almost all compared methods to fail in locating the building, which led to failed CD. However, our MGCR method not only accurately located the building at this position but also predicted the precise shape of the difference region nearly completely. In sample C, MGCR's prediction was closest to the GT, with only minor errors along the edges of the predicted target area, while other methods produced blurred localizations and considerable fragmentation in detecting the difference region in the upper right. For samples E and F, there were parts of detected difference regions that differed significantly from actual change areas; these false positives often resembled true difference areas but were nonbuilding regions. Methods like ChangeCLIP, SEIFNet, and AERNet exhibited redundant predictions in these areas. As shown in the Table I, we conducted comparisons across multiple mainstream methods using four common evaluation metrics corresponding to the visualized results. Our method outperformed the second-best, ChangeCLIP, with improvements of 0.18% in $F1$ and 0.29% in IoU.

TABLE I

INDICATOR RESULTS FOR THE LEVIR-CD DATASET. RED COLOR REPRESENTS THE BEST RESULTS AND BLUE COLOR REPRESENTS THE SECOND BEST RESULTS (%)

Methods	F1	IoU	Precision	Recall
SNUNet [39]	88.16	78.83	89.18	87.17
BIT [40]	89.31	80.68	89.24	89.37
DTCDSCN [41]	87.67	78.05	88.53	86.83
ChangeFormer [42]	90.40	82.48	92.05	88.80
ICIF-Net [43]	91.18	83.85	91.13	90.57
DMINet [44]	90.71	82.99	92.52	89.95
AERNet [45]	90.78	83.11	89.97	91.59
SEIFNet [46]	90.86	83.25	94.29	87.67
ChangeCLIP [47]	91.89	85.01	92.93	90.89
Ours	92.07	85.30	92.52	91.22

As shown in Fig. 5, we selected representative samples A–F from the WHU-CD dataset based on characteristics such as building quantity and density, considering diverse perspectives. We provide detailed analysis of several samples. In sample A, the bitemporal images depict structural renovations at corresponding locations, resulting in a relatively complex form of building change. In contrast, our method, along with DMINet, delivered relatively good performance. Sample B

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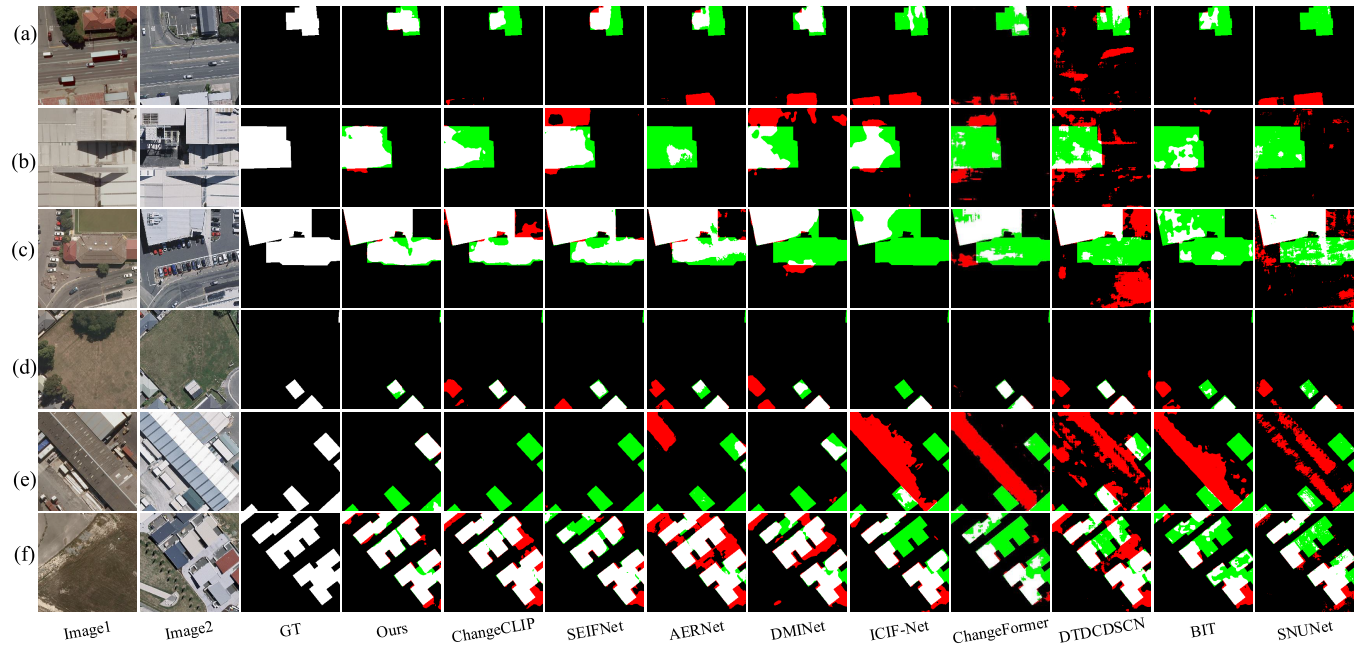


Fig. 5. Comparison of MGCR-Net with other methods, where (a)–(f) are representative examples from the WHU-CD dataset. In this figure, the green areas represent the predicted parts that are missing compared to the GT, while the red areas represent the redundant predicted parts compared to GT. The combination of the PVT and CLIP visual-text encoders forms a powerful model architecture, significantly improving both the accuracy of target area segmentation after building change detection and the precision of boundary segmentation.

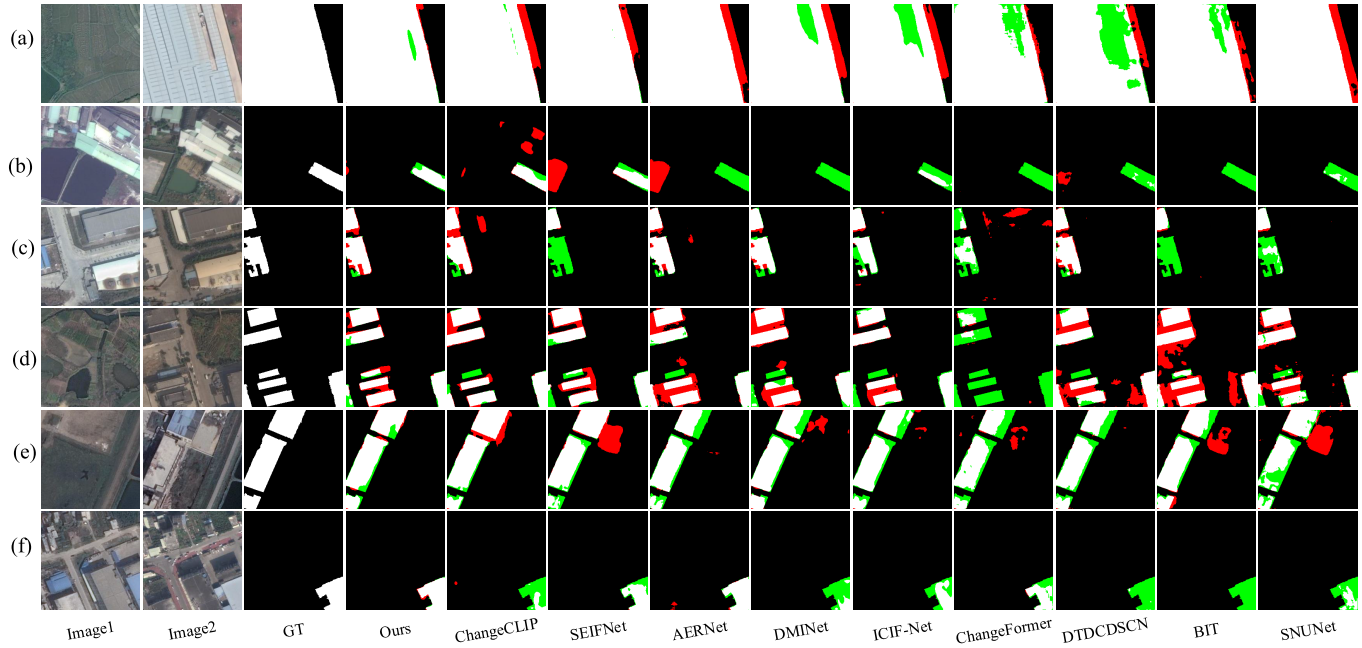


Fig. 6. Comparison of MGCR-Net with other methods, where (a)–(f) are representative examples from the GZ-CD dataset. In this figure, the green areas represent the predicted parts that are missing compared to the GT, while the red areas represent the redundant predicted parts compared to GT. The combination of SGCM and LViT effectively suppresses local pseudo-change noise and significantly reduces various types of redundant interference.

presents a similar issue, where variations in building height combined with shadows introduced substantial interference. While SEIFNet predicted the main change region well, it also produced excessive false positives. Our method maintained strong robustness under these conditions. Sample D illustrates a typical case where trees obscure buildings, causing models to misinterpret the area as containing only vegetation, which resulted in incorrect predictions. ChangeCLIP, AERNet, and DMINet all failed in this scenario. However, our MGCR successfully overcame this challenge, demonstrating strong generalization capabilities when handling complex or occluded situations. As presented in the Table II, we compared our model with multiple mainstream methods using four standard evaluation metrics. Our MGCR outperformed the second-best method, ChangeCLIP, by 0.51% in $F1$ and 0.92% in IoU.

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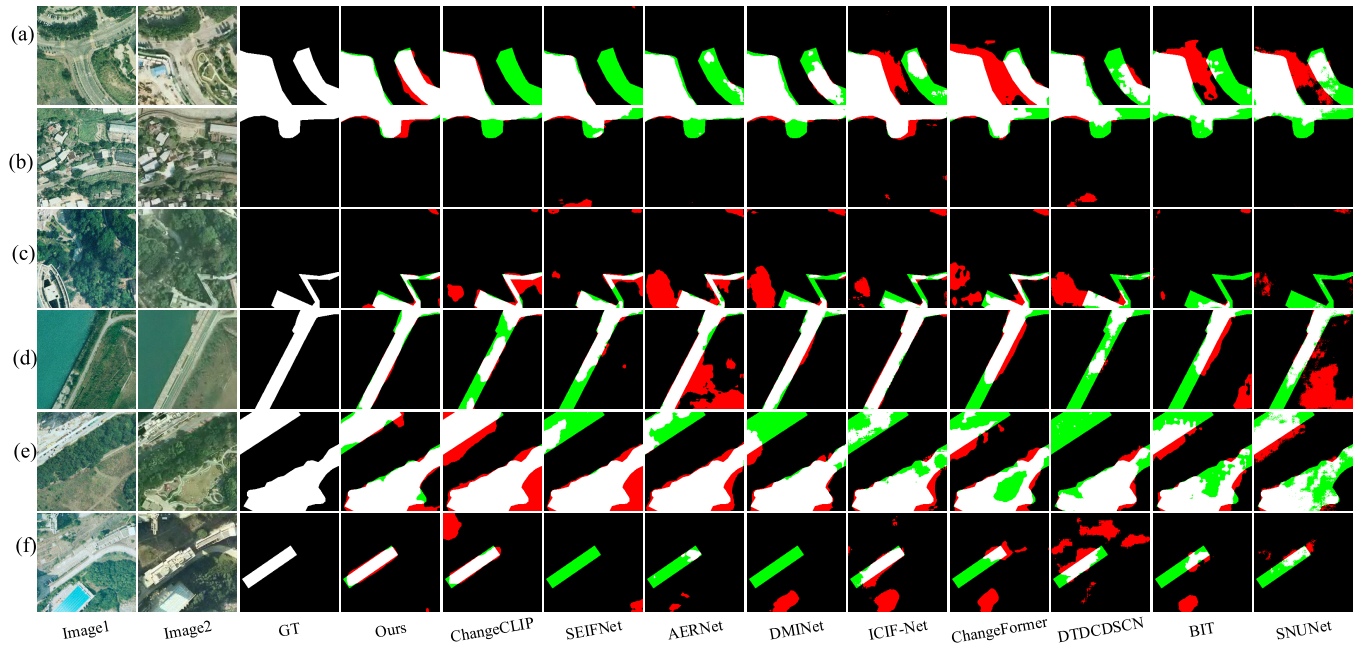


Fig. 7. Comparison of MGCR-Net with other methods, where (a)–(f) are representative examples from the SYSU-CD dataset. In this figure, the green areas represent the predicted parts that are missing compared to the GT, while the red areas represent the redundant predicted parts compared to GT. MGCR demonstrates strong generalization capability even when applied to change detection datasets that are not limited to building areas. When dealing with large-scale divergent noise, the model exhibits excellent stability and robustness.

TABLE II

INDICATOR RESULTS FOR THE WHU-CD DATASET. RED COLOR REPRESENTS THE BEST RESULTS AND BLUE COLOR REPRESENTS THE SECOND BEST RESULTS (%)

Methods	F1	IoU	Precision	Recall
SNUNet [39]	88.34	79.11	91.34	85.53
BIT [40]	87.47	77.73	88.71	86.27
DTCDSCN [41]	90.48	82.62	91.84	89.16
ChangeFormer [42]	86.88	76.81	88.50	85.33
ICIF-Net [43]	90.77	83.09	92.93	88.70
DMINet [44]	91.49	84.31	92.65	90.35
AERNet [45]	92.18	85.49	92.47	91.89
SEIFNet [46]	93.29	87.43	93.99	92.61
ChangeCLIP [47]	94.40	89.40	95.18	93.64
Ours	94.91	90.32	96.42	93.45

TABLE III

INDICATOR RESULTS FOR THE GZ-CD DATASET. RED COLOR REPRESENTS THE BEST RESULTS AND BLUE COLOR REPRESENTS THE SECOND BEST RESULTS (%)

Methods	F1	IoU	Precision	Recall
SNUNet [39]	84.25	72.79	84.25	81.82
BIT [40]	80.23	66.99	82.40	78.18
DTCDSCN [41]	83.00	70.93	88.19	78.38
ChangeFormer [42]	73.66	58.30	84.59	65.23
ICIF-Net [43]	85.09	74.05	89.90	80.76
DMINet [44]	81.98	69.46	87.92	76.79
AERNet [45]	84.42	73.03	88.06	81.07
SEIFNet [46]	87.48	77.75	89.64	85.43
ChangeCLIP [47]	88.38	79.18	91.86	85.15
Ours	89.55	81.07	92.07	87.16

As shown in Fig. 6, we selected representative samples A– from the GZ-CD dataset based on building quantity and density, highlighting different perspectives. Several samples are discussed in detail below. In sample A, while the overall prediction accuracy for the large building area was high, there were noticeable inconsistencies along the building edges compared to the main structure, which affected the prediction results to some extent. Except for ChangeFormer, other mainstream methods produced stripe-like redundant predictions, indicating that our method performs excellently when dealing with edge-related differences. Samples C, D, and E demonstrate the model's performance in detecting changes in complex urban scenes with multistructured building clusters. Most comparison methods suffered from issues such as buildings being mistakenly merged or individual structures being omitted. In contrast, our method showed strong adaptability and robustness in capturing such structural changes. As shown in the Table III, we compared our method with several mainstream models using four commonly used evaluation metrics. Our MGCR achieved the best results, outperforming the second-best method, ChangeCLIP, by 1.17% in $F1$ and 1.89% in IoU.

As shown in Fig. 7, we selected representative samples A–F from the SYSU-CD dataset, which includes diverse change

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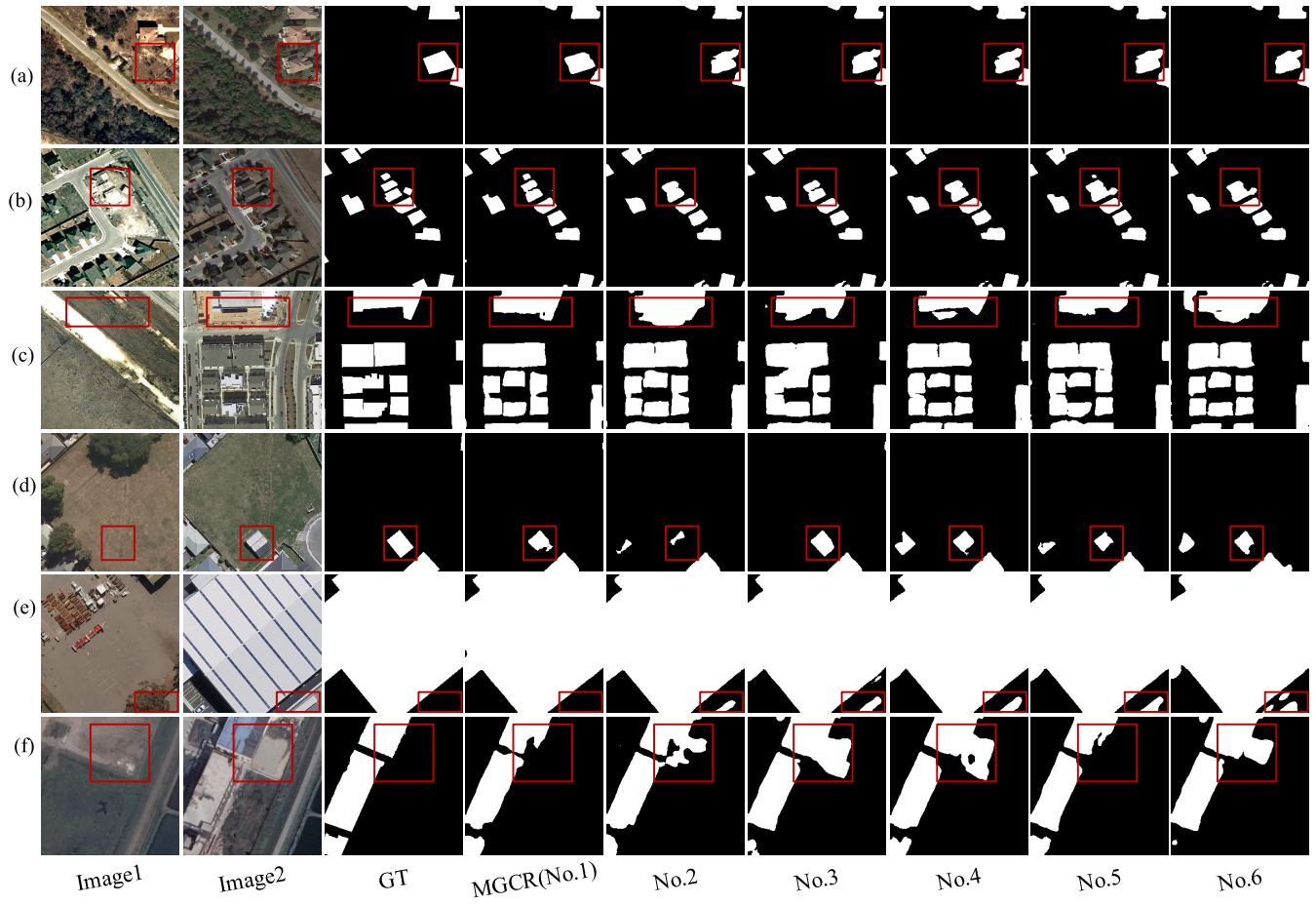


Fig. 8. Ablation experiments of MGCR-Net with other methods, where (a)–(f) are representative examples from four datasets, with key focus areas in the samples highlighted using red boxes.

TABLE IV
INDICATOR RESULTS FOR THE SYSU-CD DATASET. RED COLOR REPRESENTS THE BEST RESULTS AND BLUE COLOR REPRESENTS THE SECOND-BEST RESULTS (%)

Methods	F1	IoU	Precision	Recall
SNUNet [39]	73.14	57.66	74.09	72.21
BIT [40]	74.32	59.13	81.75	68.12
DTCDSCN [41]	77.22	62.89	80.42	74.26
ChangeFormer [42]	74.41	59.24	77.99	71.14
ICIF-Net [43]	76.08	61.40	79.16	73.24
DMINet [44]	80.06	66.75	84.93	75.72
AERNet [45]	80.26	67.02	82.50	78.13
SEIFNet [46]	81.46	68.72	83.59	79.44
ChangeCLIP [47]	82.61	70.38	86.64	78.94
Ours	82.91	70.80	86.52	79.58

regions such as building clusters and roads. Below, we provide detailed explanations for selected samples. In sample A, the two temporal images represent vegetation coverage and land use, respectively. Due to the discontinuity in vegetation in the right area, several mainstream methods failed to detect changes completely. Leveraging our novel VL alignment and reconstruction strategy, we achieved a complete and accurate prediction of the change area. In sample C, the paired images reflect a change from vegetation to road construction. Competing methods exhibited either large redundant predictions or significant omissions in road detection. Although our method showed some disconnection in the predicted road changes, it effectively captured semantic information, maintaining the correct shape and structural consistency. In sample E, compared to advanced methods like ChangeCLIP and SEIFNet, our method successfully avoided the common tendency of over-predicting large-scale change areas. Instead, it performed precise segmentation without excessive redundancy, demonstrating improved adaptability in complex scenarios. As shown in the Table IV, we compared our method against several leading approaches using four standard evaluation metrics. Our method outperformed the second-best approach, ChangeCLIP, with an improvement of 0.3% in $F1$ and 0.42% IoU.

Overall, we conducted extensive comparisons of MGCR with mainstream methods on four public datasets. Visual analysis of representative change scenarios and detailed quantitative results demonstrate the effectiveness of MGCR.

E. Ablation Study

To validate the effectiveness of our MGCR method, we conducted ablation studies from multiple perspectives, including adjustments to loss coefficients and the removal of individual

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TABLE V
OVERVIEW OF ABLATION RESULTS FOR MGCR-NET AND METRIC OUTCOMES ON FOUR DATASETS (%)

No.	SGCM-L	SGCM-V	LViT	LEVIR-CD		WHU-CD		GZ-CD		SYSU-CD	
				F1	IoU	F1	IoU	F1	IoU	F1	IoU
1	✓	✓	✓	**92.07**/91.91	**85.30**/85.03	**94.91**/94.65	**90.32**/89.85	**89.55**/89.35	**81.07**/80.74	**82.91**/82.76	**70.80**/70.60
2	✓	×	✓	91.77/91.75	84.79/84.76	94.60/94.43	89.76/89.45	88.83/88.70	79.91/79.69	82.50/82.33	70.21/69.96
3	×	✓	✓	91.67/91.73	84.62/84.72	94.53/94.46	89.62/89.50	89.04/88.82	80.25/79.88	82.55/82.43	70.29/70.11
4	×	×	✓	91.58/91.47	84.47/84.28	94.38/94.24	89.36/89.10	88.38/88.35	79.18/79.13	82.19/82.02	69.77/69.52
5	✓	✓	×	91.58/-	84.47/-	94.28/-	89.18/-	88.72/-	79.72/-	82.27/-	69.88/-
6	×	×	×	91.25/-	83.90/-	93.87/-	88.44/-	88.06/-	78.67/-	81.76/-	69.15/-

modules. Specifically, we set the loss coefficients to 0.8, 0.1, 0.1 and 1, 0, 0, respectively, for key parameter sensitivity analysis. As shown in the Table V, the values before "/" represent the configuration of 0.8, 0.1, 0.1, while the values after "/" represent the configuration of 1, 0, 0. According to the experimental data, the former demonstrates greater stability and superiority compared to the latter. For the SGCM module, we configured SGCM-V with only visual-conditioned reconstruction and SGCM-L with only language-conditioned reconstruction to investigate the individual effects of visual and language reconstruction. Additionally, we performed ablation on the LViT module. Since removing the LViT module results in the inability to form two branches necessary for loss comparison, it is denoted as a blank "–".

As shown in Fig. 8, we selected representative samples from multiple datasets to visualize the ablation experiment results under the loss coefficient setting of 0.8, 0.1, 0.1. In sample A, the extended parts with similar color and texture to the buildings were predicted as change areas by several ablated models. In sample B, MGCR better predicted the small-scale changed buildings and clearly distinguished individual structures, while the ablated models showed varying degrees of missed detections and merged change areas. In sample C, the incomplete SGCM module led to poorer prediction results and a greater tendency for large redundant areas. In sample D, No. 3 visually demonstrates the advantage of multimodal textual data compared to No. 2 and No. 4, avoiding both redundant and missing predictions. Sample F corresponding to No. 5 reflects the overall effect of SGCM, where cross-dimensional alignment of visual and textual features enabled efficient semantic information integration.

As shown in Fig. 9, to further validate the effectiveness of the mechanisms introduced in MGCR for CD tasks, we conducted a heatmap-based visualization analysis of the model's attention regions during training. At the feature levels of 16×16 and 32×32 pixels, the attention regions of MGCR align closely with the actual change areas, indicating that the model can accurately focus on semantically significant changes. In contrast, models with the original fusion strategy produce noisy, dispersed heatmaps with greater interference. This highlights MGCR's superior focus on change areas and confirms the importance of the SGCM module and LViT architecture in improving semantic understanding and spatial awareness.

As shown in Fig. 10, we applied image tiles from real remote sensing scenes to further demonstrate the practical significance of the MGCR method. These tiles were randomly selected from complex scenarios, providing sufficient variability to test the model's generalization ability. Specifically, we used the model weights obtained from training on the LEVIR-CD dataset to evaluate the real-world scene images. Although there is no ground truth reference for these remote sensing image tiles, the visualized predictions clearly show that our model produces more structured and organized building segmentation with excellent performance. This validates the effectiveness and robustness of MGCR in handling complex change scenarios and demonstrates its practical applicability.

F. Parameters and FLOPs Comparison

As shown in Table VI, we compare multimodal graph-conditioned vision-language reconstruction network (MGCR-Net) with other methods in terms of the number of parameters and FLOPs. The number of parameters refers to the total trainable parameters in the model, reflecting its storage requirements and computational cost during training. FLOPs represent the number of floating-point operations required for a single forward pass, indicating the computational efficiency of the model. In terms of LLM parameter considerations and tradeoffs, the computational cost of ChangeCLIP is difficult to estimate precisely. Our MGCR model measures the parameter count and FLOPs of a multistructured network, where both the parameter size, FLOPs, and runtime are relatively large. However, this higher computational cost provides the advantage of stronger generalization ability and superior performance.

V. DISCUSSION

This article leverages LLaVA's capabilities in visual-language understanding and intelligent dialog interaction to construct cross-modal textual data based on raw image inputs. By integrating PVT as the visual encoder and CLIP as the text encoder, we build a text-guided cross-modal CD framework. For building CD tasks, the model must first ensure precise localization and segmentation of buildings, followed by extracting difference regions through concatenation and differencing operations. Therefore, we modify prompts to direct LLaVA to generate building-related textual descriptions, while also partially describing other foreground and background information. However, among the large amount of generated text, there remain cases of low image–text similarity. We continue to explore the application of generative AI in multimodal data generation, pushing deeper in building CD

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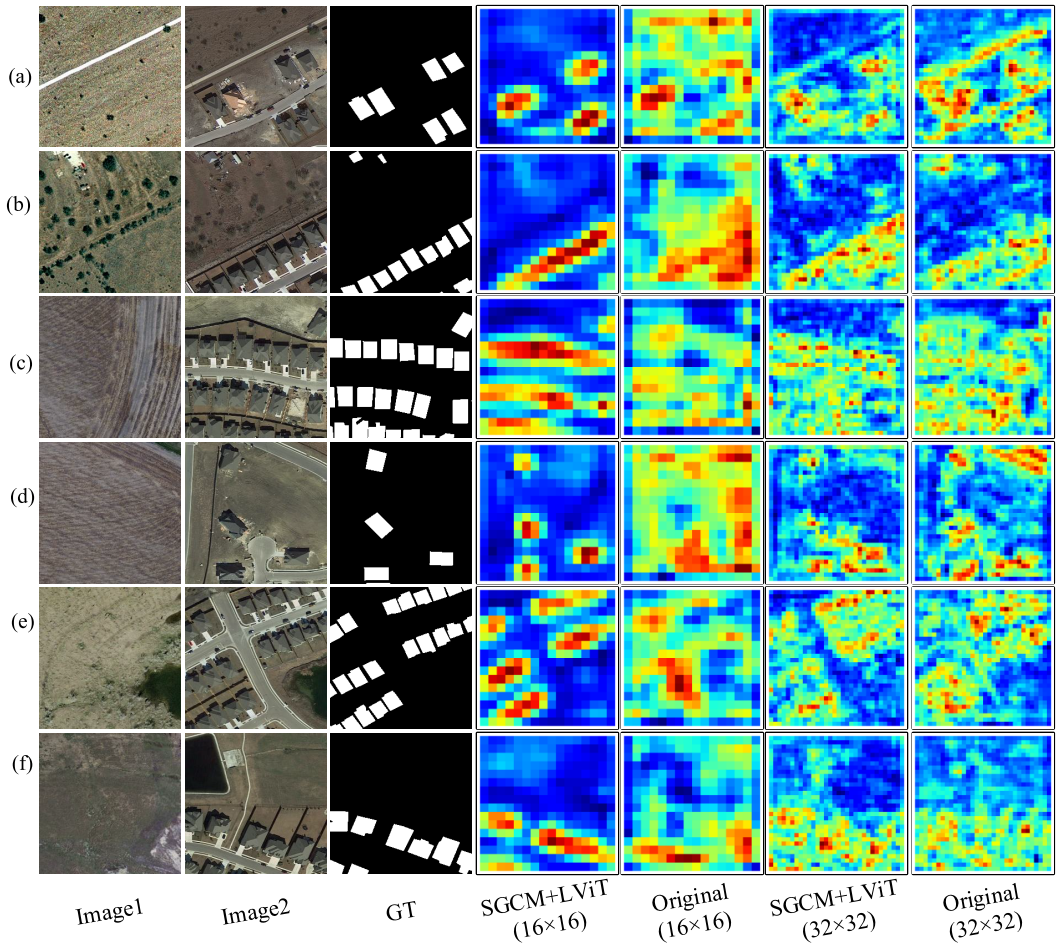


Fig. 9. Visualization of MGCR-Net, where (a)–(f) are heatmap visualization results depict the semantic fusion strategies of SGCM and LViT at different pixel locations, compared with the original fusion strategy.

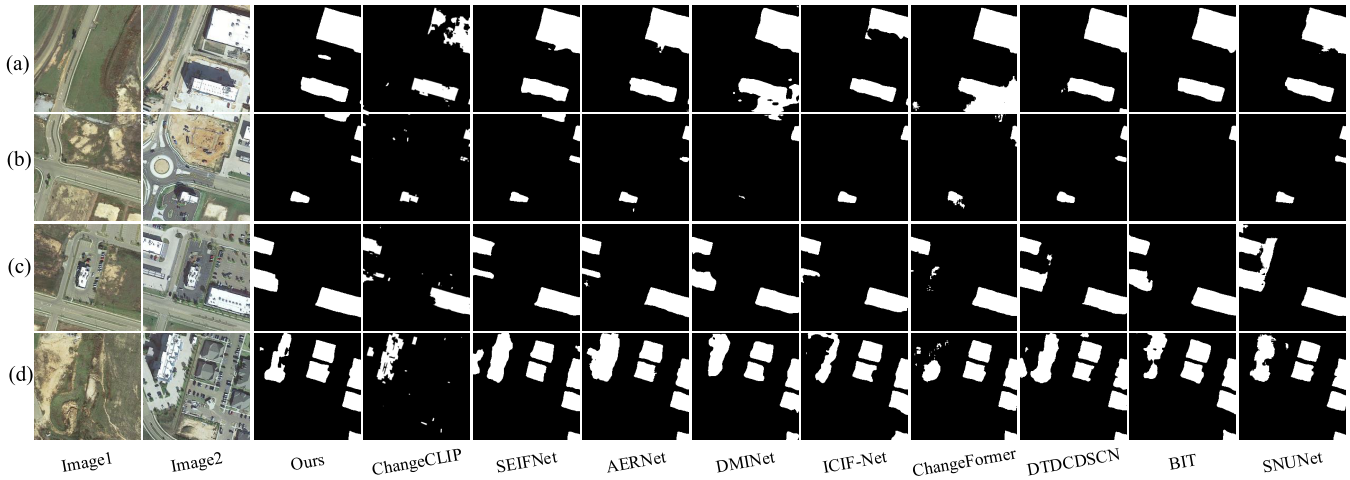


Fig. 10. Comparison of MGCR-Net with other methods, where representative examples (a)–(d) demonstrate the performance of predicting CD results in real-world scenarios. This scenario uses Google Earth images of Oxford, Mississippi (2012–2019).

tasks to generate textual data with higher image–text matching degrees. At the same time, we aim to broaden the scope in CD tasks to generate textual data that is not limited to buildings but can describe specific and intuitive change targets.

In addition, we analyzed potential optimization strategies for real-time deployment of the model and provided a more detailed discussion of its limitations. Real-time deployment of MLLMs remains a significant challenge; however, our model implements pruning-based optimization for multimodal text data generation, significantly reducing redundant computations when processing textual information. While our approach can mitigate the effects of global noise in traditional CD tasks, it

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TABLE VI
COMPARISON OF PARAMETERS(M) AND FLOPS(G) ALONG WITH A BRIEF DESCRIPTION OF THE COMPARISON METHODS FOR $F1(\%)$ AND IOU(%) PERFORMANCE

Methods	Params(M)	FLOPs(G)	LEVIR-CD		WHU-CD		GZ-CD		SYSU-CD	
			F1	IoU	F1	IoU	F1	IoU	F1	IoU
SNUNet [39]	12.03	54.88	88.16	78.83	88.34	79.11	84.25	72.79	73.14	57.66
BIT [40]	3.55	10.59	89.31	80.68	87.47	77.73	80.23	66.99	74.32	59.13
DTCDSCN [41]	41.07	13.21	87.67	78.05	90.48	82.62	83.00	70.93	77.22	62.89
ChangeFormer [42]	41.03	202.83	90.40	82.48	86.88	76.81	73.66	58.30	74.41	59.24
ICIF-Net [43]	25.83	25.27	91.18	83.85	90.77	83.09	85.09	74.05	76.08	61.40
DMINet [44]	6.24	14.55	90.71	82.99	91.49	84.31	81.98	69.46	80.06	66.75
AERNet [45]	25.36	12.82	90.78	83.11	92.18	85.49	84.42	73.03	80.26	67.02
SEIFNet [46]	8.37	27.9	90.86	83.25	93.29	87.43	87.48	77.75	81.46	68.72
ChangeCLIP [47]	-	-	91.89	85.01	94.40	89.40	88.38	79.18	82.61	70.38
Ours	77.71	19.31	**92.07**	**85.30**	**94.91**	**90.32**	**89.55**	**81.07**	**82.91**	**70.80**

still has limitations in addressing inherent challenges such as targets being heavily occluded. Therefore, in future work, we will focus on tackling strong dynamic change interference and severe occlusion issues for MLLMs under real-time deployment.

VI. CONCLUSION

In this article, we proposed a MGCR-Net for RSCD. We utilized MLLM to generate cross-modal textual data, which guided the model to achieve multimodal input and multimodal interaction, thereby enhancing the model's ability to perceive change regions in remote sensing images from multiple perspectives. PVT and CLIP were employed as feature encoders to process image and text information, respectively. Furthermore, we proposed SGCM for the first time to align image–text features, SGCM constructed VL tokens via a graph structure, and combined multihead attention to perform both image and text reconstruction, achieving fully interactive semantic image–text feature information. Subsequently, LViT was used to fuse the reconstructed image–text features, during which the mean-squared error between image features and image–text features was calculated to correct feature discrepancies. This, together with binary cross-entropy loss, jointly guided model optimization. Overall, we optimized the strategy for generating textual data via MLLM, processed multimodal features through feature encoders, designed SGCM and leveraged LViT for image–text feature alignment and fusion, and established relevant losses to guide multimodal model training. Comprehensive experiments on multiple public datasets validated the effectiveness of the MGCR multimodal architecture, with results demonstrating that MGCR achieved outstanding performance compared to mainstream methods.

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